



Chapter 8: Complex Data Types

Database System Concepts, 7th Ed.

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Outline

- Semi-Structured Data
- Object Orientation
- Textual Data
- Spatial Data



Semi-Structured Data

- Many applications require storage of complex data, whose schema changes often
- The relational model's requirement of atomic data types may be an overkill
 - E.g. storing set of interests as a set-valued attribute of a user profile may be simpler than normalizing it
- Data exchange can benefit greatly from semi-structured data
 - Exchange can be between applications, or between back-end and front-end of an application
 - Web-services are widely used today, with complex data fetched to the front-end and displayed using a mobile app or JavaScript
- JSON and XML are widely used semi-structured data models



Features of Semi-Structured Data Models

- **Flexible schema**
 - Profile** : { **Name:** {}, **Interests:** {**Reading, Travelling,,**}}
 - Profile** : { **Name:** {}, }
 - **Wide column** representation: allow each tuple to have a different set of attributes, can add new attributes at any time
 - **Sparse column** representation: schema has a fixed but large set of attributes, by each tuple may store only a subset
- **Multivalued data types**
 - Profile** : { **Name:** {}, **Interests:** {**Reading, Travelling,,**}}
 - Profile** : { **Name:** {}, **Interests:** {} }
 - **Sets, multisets**
 - E.g.: set of interests {'basketball', 'La Liga', 'cooking', 'anime', 'jazz'}
 - **Key-value map** (or just **map** for short)
 - Store a set of key-value pairs
 - E.g. {(brand, Apple), (ID, MacBook Air), (size, 13), (color, silver)}
 - Operations on maps: *put*(key, value), *get*(key), *delete*(key)
 - **Arrays**
 - Widely used for scientific and monitoring applications



Features of Semi-Structured Data Models

■ Arrays

- Widely used for scientific and monitoring applications
- E.g. readings taken at regular intervals can be represented as array of values instead of (time, value) pairs
 - [5, 8, 9, 11] instead of {(1,5), (2, 8), (3, 9), (4, 11)}

■ Multi-valued attribute types

- Modeled using *non first-normal-form (NFNF)* data model
- Supported by most database systems today

■ **Array database**: a database that provides specialized support for arrays

- E.g. compressed storage, query language extensions etc
- Oracle GeoRaster, PostGIS, SciDB, etc



Nested Data Types

- Hierarchical data is common in many applications
- JSON: JavaScript Object Notation
 - Widely used today
- XML: Extensible Markup Language
 - Earlier generation notation, still used extensively



JSON

- Textual representation widely used for data exchange
- Example of JSON data

```
{  
  "ID": "22222",  
  "name": {  
    "firstname": "Albert",  
    "lastname": "Einstein"  
  },  
  "deptname": "Physics",  
  "children": [  
    {"firstname": "Hans", "lastname": "Einstein" },  
    {"firstname": "Eduard", "lastname": "Einstein" }  
  ]  
}
```

- Types: integer, real, string, and
 - *Objects*: are key-value maps, i.e. sets of (attribute name, value) pairs
 - Arrays are also key-value maps (from offset to value)



JSON

Profile as JSON

Profile : {Interests{}, Address: {}}

- JSON is ubiquitous in data exchange today
 - Widely used for web services
 - Most modern applications are architected around on web services
- SQL extensions for
 - JSON types for storing JSON data
 - Extracting data from JSON objects using path expressions
 - E.g. *V* → *ID*, or *v.ID*
 - Generating JSON from relational data
 - E.g. `json.build_object('ID', 12345, 'name', 'Einstein')`
 - Creation of JSON collections using aggregation
 - E.g. `json_agg` aggregate function in PostgreSQL
 - Syntax varies greatly across databases
- JSON is verbose
 - Compressed representations such as BSON (Binary JSON) used for efficient data storage

**Select Name, Id, Year, Profile->Interests
from Students;**



XML

- XML uses tags to mark up text
- E.g.
 - <course>
 - <course id> CS-101 </course id>
 - <title> Intro. to Computer Science </title>
 - <dept name> Comp. Sci. </dept name>
 - <credits> 4 </credits>
 - </course>
- Tags make the data self-documenting
- Tags can be hierarchical



Example of Data in XML

```
■ <purchase order>
  <identifier> P-101 </identifier>
  <purchaser>
    <name> Cray Z. Coyote </name>
    <address> Route 66, Mesa Flats, Arizona 86047, USA
  </address>
</purchaser>
<supplier>
  <name> Acme Supplies </name>
  <address> 1 Broadway, New York, NY, USA </address>
</supplier>
<itemlist>
  <item>
    <identifier> RS1 </identifier>
    <description> Atom powered rocket sled </description>
    <quantity> 2 </quantity>
    <price> 199.95 </price>
  </item>
  <item>...</item>
</itemlist>
<total cost> 429.85 </total cost>
....
</purchase order>
```



XML Cont.

- XQuery language developed to query nested XML structures
 - Not widely used currently
- SQL extensions to support XML
 - Store XML data
 - Generate XML data from relational data
 - Extract data from XML data types
 - Path expressions
- See Chapter 30 (online) for more information



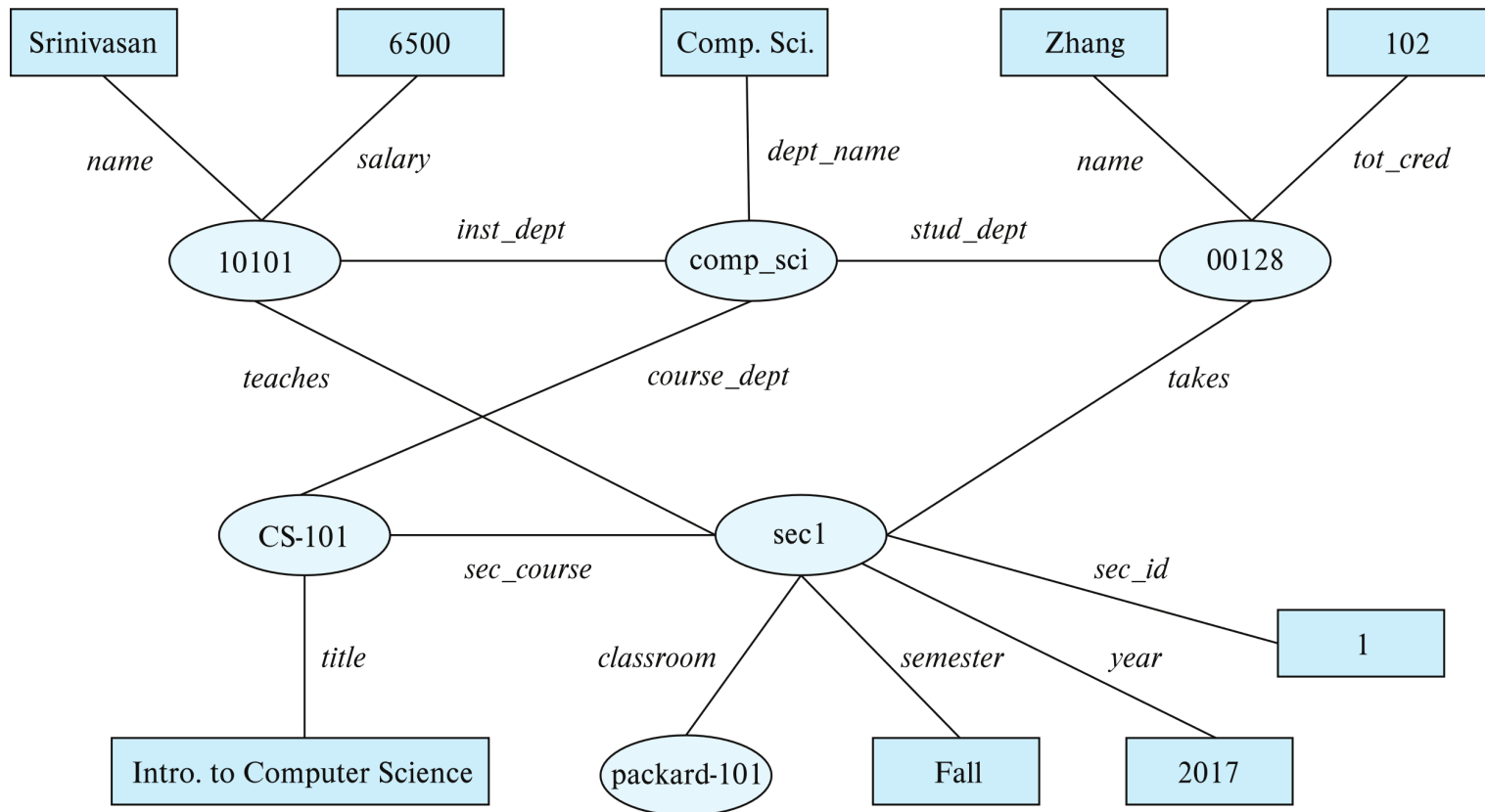
Knowledge Representation

- Representation of human knowledge is a long-standing goal of AI
 - Various representations of facts and inference rules proposed over time
- **RDF: Resource Description Format**
 - Simplified representation for facts, represented as triples (*subject, predicate, object*)
 - E.g. (NBA-2019, *winner*, Raptors)
 - (Washington-DC, *capital-of*, USA)
 - (Washington-DC, *population*, 6,200,000)
 - Models objects that have attributes, and relationships with other objects
 - Like the ER model, but with a flexible schema
 - (*ID, attribute-name, value*)
 - (*ID1, relationship-name, ID2*)
 - Has a natural graph representation



Graph View of RDF Data

- Knowledge graph





Triple View of RDF Data

10101	instance-of	instructor .
10101	name	"Srinivasan" .
10101	salary	"6500" .
00128	instance-of	student .
00128	name	"Zhang" .
00128	tot_cred	"102" .
comp_sci	instance-of	department .
comp_sci	dept_name	"Comp. Sci." .
biology	instance-of	department .
CS-101	instance-of	course .
CS-101	title	"Intro. to Computer Science" .
CS-101	course_dept	comp_sci .
sec1	instance-of	section .
sec1	sec_course	CS-101 .
sec1	sec_id	"1" .
sec1	semester	"Fall" .
sec1	year	"2017" .
sec1	classroom	packard-101 .
sec1	time_slot_id	"H" .
10101	inst_dept	comp_sci .
00128	stud_dept	comp_sci .
00128	takes	sec1 .
10101	teaches	sec1 .



Querying RDF: SPARQL

- Triple patterns
 - `?cid title "Intro. to Computer Science"`
 - `?cid title "Intro. to Computer Science"`
`?sid course ?cid`
- SPARQL queries
 - **select** `?name`
where {
 - `?cid title "Intro. to Computer Science" .`
 - `?sid course ?cid .`
 - `?id takes ?sid .`
 - `?id name ?name .`}
 - Also supports
 - Aggregation, Optional joins (similar to outerjoins), Subqueries, etc.
 - Transitive closure on paths



RDF Representation (Cont.)

- RDF triples represent binary relationships
- How to represent n-ary relationships?
 - Approach 1 (from Section 6.9.4): Create artificial entity, and link to each of the n entities
 - E.g. (Barack Obama, *president-of*, USA, 2008-2016) can be represented as
(*e1*, *person*, Barack Obama), (*e1*, *country*, USA),
(*e1*, *president-from*, 2008) (*e1*, *president-till*, 2016)
 - Approach 2: use **quads** instead of triples, with context entity
 - E.g. (Barack Obama, *president-of*, USA, *c1*)
(*c1*, *president-from*, 2008) (*c1*, *president-till*, 2016)
- RDF widely used as knowledge base representation
 - DBPedia, Yago, Freebase, WikiData, ..
- **Linked open data** project aims to connect different knowledge graphs to allow queries to span databases

Wikipedia
Dbpedia



Object Orientation

OWL 1 & 2 —Web Ontology Language

- **Object-relational data model** provides richer type system
 - with complex data types and object orientation
- Applications are often written in object-oriented programming languages
 - Type system does not match relational type system
 - Switching between imperative language and SQL is troublesome
- Approaches for integrating object-orientation with databases
 - Build an **object-relational database**, adding object-oriented features to a relational database
 - Automatically convert data between programming language model and relational model; data conversion specified by **object-relational mapping**
 - Build an **object-oriented database** that natively supports object-oriented data and direct access from programming language



Object-Relational Database Systems

- User-defined types
 - **create type** *Person*
(*ID* **varchar**(20) **primary key**,
name **varchar**(20),
address **varchar**(20)) **ref from**(*ID*); /* More on this later */
create table *people* **of** *Person*;
- Table types
 - **create type** *interest* **as table** (
topic **varchar**(20),
degree_of_interest **int**);
create table *users* (
ID **varchar**(20),
name **varchar**(20),
interests *interest*);
- Array, multiset data types also supported by many databases
 - Syntax varies by database



Type and Table Inheritance

- Type inheritance
 - **create type** *Student* **under** *Person*
(*degree* **varchar**(20)) ;
create type *Teacher* **under** *Person*
(*salary* **integer**);
- Table inheritance syntax in PostgreSQL and oracle
 - **create table** *students*
(*degree* **varchar**(20))
inherits *people*;
create table *teachers*
(*salary* **integer**)
inherits *people*;
 - **create table** *people* **of** *Person*;
create table *students* **of** *Student*
under *people*;
create table *teachers* **of** *Teacher*
under *people*;



Reference Types

- Creating reference types
 - **create type** *Person*
 (*ID* **varchar**(20) **primary key**,
 name **varchar**(20),
 address **varchar**(20))
 ref from(*ID*);
create table *people* **of** *Person*;
create type *Department* (
 dept_name **varchar**(20),
 head **ref**(*Person*) **scope** *people*);
create table *departments* **of** *Department*
insert into *departments* **values** ('CS', '12345')
 - System generated references can be retrieved using subqueries
 - (**select** **ref**(*p*) **from** *people* **as** *p* **where** *ID* = '12345')
- Using references in **path expressions**
 - **select** *head->name*, *head->address*
from *departments*;



Object-Relational Mapping

Student [] students;
students.add(new Student())—>INSERT into

- Object-relational mapping (ORM) systems allow
 - Specification of mapping between programming language objects and database tuples
 - Automatic creation of database tuples upon creation of objects
 - Automatic update/delete of database tuples when objects are update/deleted
 - Interface to retrieve objects satisfying specified conditions
 - Tuples in database are queried, and object created from the tuples
- Details in Section 9.6.2
 - Hibernate ORM for Java
 - Django ORM for Python



Textual Data

- **Information retrieval:** querying of unstructured data
 - Simple model of keyword queries: given query keywords, retrieve documents containing all the keywords
 - More advanced models rank relevance of documents
 - Today, keyword queries return many types of information as answers
 - E.g. a query “cricket” typically returns information about ongoing cricket matches
- Relevance ranking
 - Essential since there are usually many documents matching keywords



Ranking using TF-IDF

- Term: keyword occurring in a document/query
- **Term Frequency:** $TF(d, t)$, the relevance of a term t to a document d
 - One definition: $TF(d, t) = \log(1 + n(d, t)/n(d))$
where $n(d, t)$ = number of occurrences of term t in document d
and $n(d)$ = number of terms in document d
- **Inverse document frequency:** $IDF(t)$
 - One definition: $IDF(t) = 1/n(t)$
- **Relevance** of a document d to a set of terms Q
 - One definition: $r(d, Q) = \sum_{t \in Q} TF(d, t) * IDF(t)$
 - Other definitions
 - take **proximity** of words into account
 - **Stop words** are often ignored



Ranking Using Hyperlinks

- Hyperlinks provide very important clues to importance
- Google introduced PageRank, a measure of popularity/importance based on hyperlinks to pages
 - Pages hyperlinked from many pages should have higher PageRank
 - Pages hyperlinked from pages with higher PageRank should have higher PageRank
 - Formalized by **random walk** model
- Let $T[i, j]$ be the probability that a random walker who is on page i will click on the link to page j
 - Assuming all links are equal, $T[i, j] = 1/N_i$
- Then PageRank[j] for each page j can be defined as
 - $P[j] = \delta N + (1 - \delta) * \sum_{i=1}^N (T[i, j] * P[i])$
 - Where N = total number of pages, and δ a constant usually set to 0.15
-



Ranking Using Hyperlinks

- Definition of PageRank is circular, but can be solved as a set of linear equations
 - Simple iterative technique works well
 - Initialize all $P[i] = 1/N$
 - In each iteration use equation $P[j] = \delta/N + (1 - \delta) * \sum_{i=1}^N (T[i, j] * P[i])$ to update P
 - Stop iteration when changes are small, or some limit (say 30 iterations) is reached.
- Other measures of relevance are also important. For example:
 - Keywords in anchor text
 - Number of times who ask a query click on a link if it is returned as an answer



Retrieval Effectiveness

- Measures of effectiveness
 - **Precision**: what percentage of returned results are actually relevant
 - **Recall**: what percentage of relevant results were returned
 - At some number of answers, e.g. precision@10, recall@10
- Keyword querying on structured data and knowledge bases
 - Useful if users don't know schema, or there is no predefined schema
 - Can represent data as graphs
 - Keywords match tuples
 - Keyword search returns closely connected tuples that contain keywords
 - E.g. on our university database given query “Zhang Katz”, Zhang matches a student, Katz an instructor and advisor relationship links them



SPATIAL DATA



Spatial Data

- Spatial databases store information related to spatial locations, and support efficient storage, indexing and querying of spatial data.
 - **Geographic data** -- road maps, land-use maps, topographic elevation maps, political maps showing boundaries, land-ownership maps, and so on.
 - **Geographic information systems** are special-purpose databases tailored for storing geographic data.
 - Round-earth coordinate system may be used
 - (Latitude, longitude, elevation)
 - **Geometric data:** design information about how objects are constructed . For example, designs of buildings, aircraft, layouts of integrated-circuits.
 - 2 or 3 dimensional Euclidean space with (X, Y, Z) coordinates



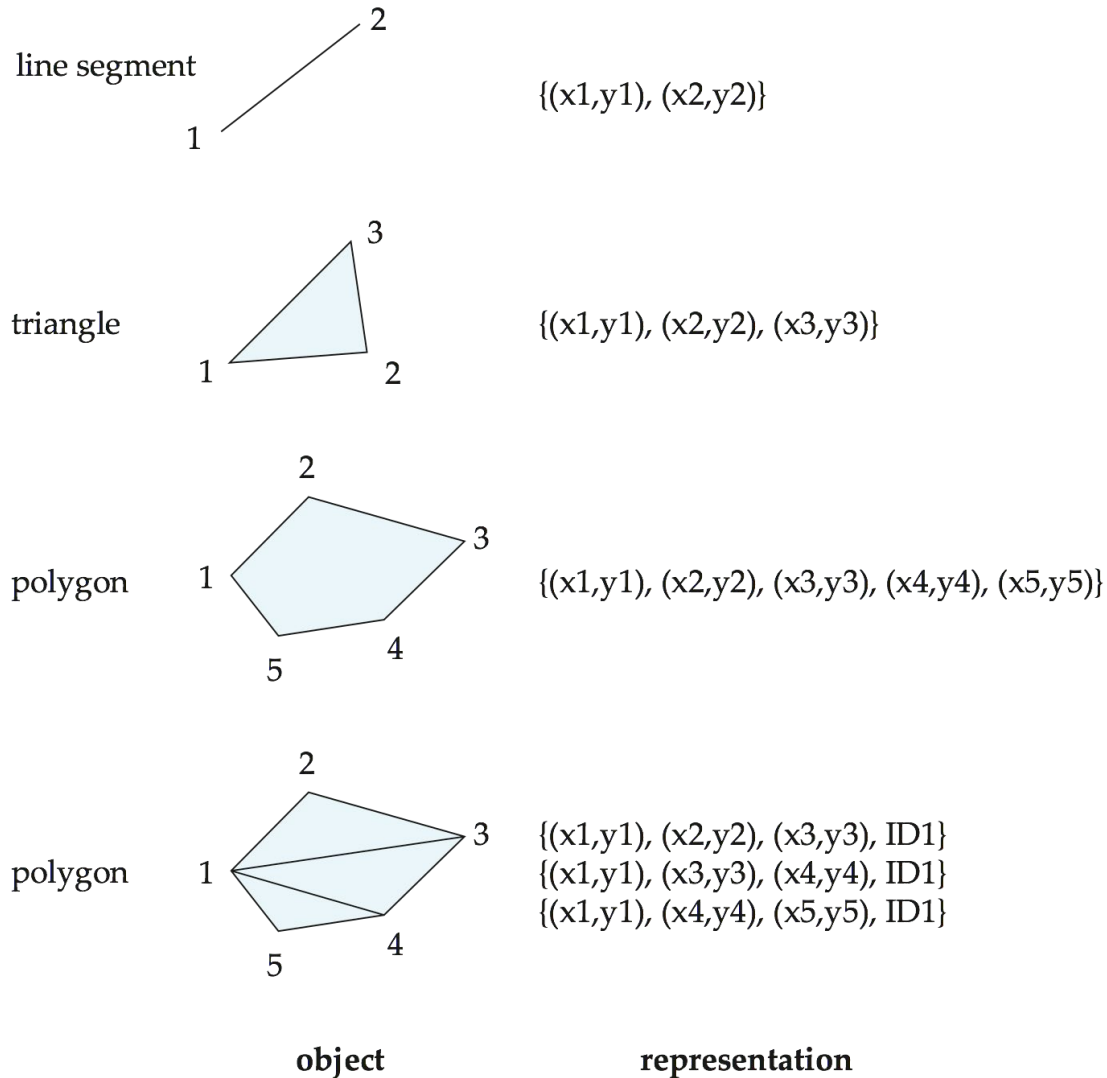
Represented of Geometric Information

Various geometric constructs can be represented in a database in a normalized fashion (see next slide)

- A **line segment** can be represented by the coordinates of its endpoints.
- A **polyline** or **linestring** consists of a connected sequence of line segments and can be represented by a list containing the coordinates of the endpoints of the segments, in sequence.
 - Approximate a curve by partitioning it into a sequence of segments
 - Useful for two-dimensional features such as roads.
 - Some systems also support *circular arcs* as primitives, allowing curves to be represented as sequences of arc
- **Polygons** is represented by a list of vertices in order.
 - The list of vertices specifies the boundary of a polygonal region.
 - Can also be represented as a set of triangles (**triangulation**)



Representation of Geometric Constructs





Representation of Geometric Information (Cont.)

- Representation of points and line segment in 3-D similar to 2-D, except that points have an extra z component
- Represent arbitrary polyhedra by dividing them into tetrahedrons, like triangulating polygons.
- Alternative: List their faces, each of which is a polygon, along with an indication of which side of the face is inside the polyhedron.
- Geometry and geography data types supported by many databases
 - E.g. SQL Server and PostGIS
 - point, linestring, curve, polygons
 - Collections: multipoint, multilinestring, multicurve, multipolygon
 - LINESTRING(1 1, 2 3, 4 4)
 - POLYGON((1 1, 2 3, 4 4, 1 1))
 - Type conversions: *ST GeometryFromText()* and *ST GeographyFromText()*
 - Operations: *ST Union()*, *ST Intersection()*, ...



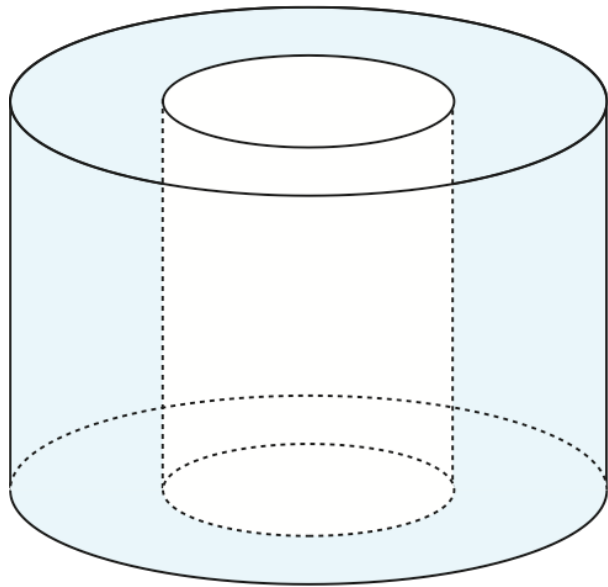
Design Databases

- Represent design components as objects (generally geometric objects); the connections between the objects indicate how the design is structured.
- Simple two-dimensional objects: points, lines, triangles, rectangles, polygons.
- Complex two-dimensional objects: formed from simple objects via union, intersection, and difference operations.
- Complex three-dimensional objects: formed from simpler objects such as spheres, cylinders, and cuboids, by union, intersection, and difference operations.
- Wireframe models represent three-dimensional surfaces as a set of simpler objects.

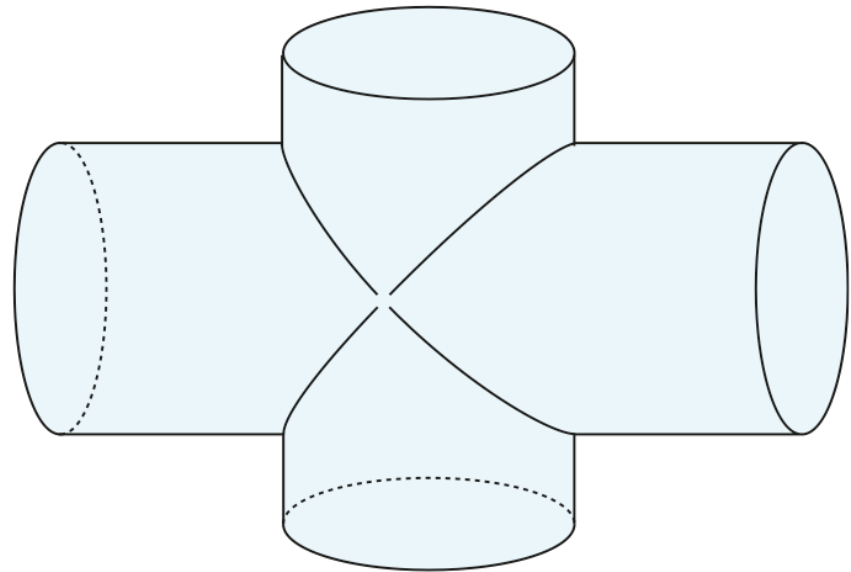


Representation of Geometric Constructs

- Design databases also store non-spatial information about objects (e.g., construction material, color, etc.)
- Spatial integrity constraints are important.
 - E.g., pipes should not intersect, wires should not be too close to each other, etc.



(a) Difference of cylinders



(b) Union of cylinders



Geographic Data

- **Raster data** consist of bit maps or pixel maps, in two or more dimensions.
 - Example 2-D raster image: satellite image of cloud cover, where each pixel stores the cloud visibility in a particular area.
 - Additional dimensions might include the temperature at different altitudes at different regions, or measurements taken at different points in time.
- Design databases generally do not store raster data.



Geographic Data (Cont.)

- **Vector data** are constructed from basic geometric objects: points, line segments, triangles, and other polygons in two dimensions, and cylinders, spheres, cuboids, and other polyhedrons in three dimensions.
- Vector format often used to represent map data.
 - Roads can be considered as two-dimensional and represented by lines and curves.
 - Some features, such as rivers, may be represented either as complex curves or as complex polygons, depending on whether their width is relevant.
 - Features such as regions and lakes can be depicted as polygons.



Spatial Queries

- **Region queries** deal with spatial regions. e.g., ask for objects that lie partially or fully inside a specified region
 - E.g. PostGIS *ST_Contains()*, *ST_Overlaps()*, ...
- **Nearness queries** request objects that lie near a specified location.
- **Nearest neighbor queries**, given a point or an object, find the nearest object that satisfies given conditions.
- **Spatial graph queries** request information based on spatial graphs
 - E.g. shortest path between two points via a road network
- **Spatial join** of two spatial relations with the location playing the role of join attribute.
- Queries that compute intersections or **unions** of regions



End of Chapter 8